

The java.lang Package

The `java.lang` package is one of the core packages in Java, automatically imported into every Java program. It provides fundamental classes and interfaces that form the foundation of the Java programming language, such as classes for primitive types, strings, threads, exceptions, and system operations.

Key Features of `java.lang`

1. No Need for Import:

- The `java.lang` package is automatically imported into every Java program.
- Example:

```
public class Example {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!"); // System is part of  
        java.lang  
    }  
}
```

2. Fundamental Classes:

- Provides classes essential for the basic operation of a Java program.
- Examples: `Object`, `String`, `Math`, `Thread`.

3. Core Interfaces:

- Includes interfaces like `Comparable` and `Runnable`.

4. Utility and Helper Methods:

- Provides methods for mathematical calculations, string manipulation, and system-level operations.
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Classes in `java.lang`

1. `Object` Class

- The root class of the Java class hierarchy.
- Every class in Java implicitly inherits from `Object`.
- Common methods:
 - `toString()`: Returns a string representation of the object.

- `equals(Object obj)` : Compares two objects for equality.
- `hashCode()` : Returns a hash code for the object.
- `clone()` : Creates a copy of the object.

2. String Class

- Represents immutable sequences of characters.
- Common methods:
 - `length()` : Returns the length of the string.
 - `charAt(int index)` : Returns the character at the specified index.
 - `substring(int start, int end)` : Extracts a portion of the string.
 - `toUpperCase()` , `toLowerCase()` : Converts the string to uppercase or lowercase.

3. StringBuilder and StringBuffer Classes

- Used for mutable string operations.
- `StringBuilder` is faster but not thread-safe.
- `StringBuffer` is thread-safe.

4. Math Class

- Provides methods for mathematical operations.
- Common methods:
 - `abs()` : Returns the absolute value.
 - `sqrt()` : Returns the square root.
 - `pow(double a, double b)` : Computes a^b .
 - `max(a, b)` , `min(a, b)` : Finds the maximum or minimum of two values.

5. System Class

- Provides system-level utilities.
- Common methods:
 - `System.out.println()` : Prints data to the console.
 - `System.currentTimeMillis()` : Returns the current time in milliseconds.
 - `System.gc()` : Suggests garbage collection.
 - `System.exit(int status)` : Exits the program.

6. Thread Class

- Represents a thread of execution.
- Common methods:
 - `start()` : Starts the thread.
 - `run()` : Contains the code to execute in the thread.

- `sleep(milliseconds)` : Causes the thread to pause for a specific time.

Key Points to Remember

1. Automatic Import:

- No need to write `import java.lang.*;` as it is imported by default.

2. Fundamental Classes:

- Essential classes like `String`, `Object`, `Math`, `System`, and `Thread` are part of `java.lang`.

3. Immutable vs Mutable Classes:

- `String` is immutable, while `StringBuilder` and `StringBuffer` are mutable.

4. Thread Support:

- Provides classes and interfaces for multithreading (`Thread` and `Runnable`).

5. Error and Exception Handling:

- `Throwable` is the parent class for all exceptions and errors.

The `java.lang` package is the backbone of Java programming, offering essential tools and classes for writing robust and efficient Java applications.